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e.g. $X = \{X_k\}_{k=0}^{\infty}$ $\quad X_k \to k \times m$ matrix

$$\|X\|_\infty = \sup_i \|X_i\|$$

Inner product of vectors defined in $\mathbb{C}^n$: $\langle x, y\rangle := x^* y \quad \forall\, x, y \in \mathbb{C}^n$

<u>Generalization:</u>

Let $\mathcal{V}$ be a vector space over $\mathbb{C}$. An inner product on $\mathcal{V}$ is a complex valued function

$$\langle \cdot, \cdot \rangle : \mathcal{V} \times \mathcal{V} \longrightarrow \mathbb{C}$$

such that

(i) $\langle x, \alpha y + \beta z\rangle = \alpha\langle x, y\rangle + \beta\langle x, z\rangle \qquad \forall\, x, y, z \in \mathcal{V}$

(ii) $\langle x, y\rangle = \overline{\langle y, x\rangle} \qquad \alpha, \beta, c \in \mathbb{C}$

(iii) $\langle x, x\rangle > 0$ if $x \neq 0$

(iv) $\langle cx, y\rangle = c\langle x, y\rangle$

A vector space with an inner product is called an inner product space.

* An innerproduct induces a norm: $\|x\| := \sqrt{\langle x, x\rangle}$
* The distance between vectors x and y: $d(x, y) := \|x - y\| = \sqrt{\langle x-y, x-y\rangle}$
* Two vectors x & y are said to be orthogonal if $\langle x, y\rangle = 0$
* A vector x is said to be orthogonal to $S \subset \mathcal{V}$ if $\langle x, y\rangle = 0 \;\forall\, y \in S$.

e.g. $\underbrace{\mathcal{C}[a, b]}$ $\quad \langle f, g\rangle = \int_a^b f^*(t)\, g(t)\, dt$ $\quad$ Norm induced $\|x\| = \sqrt{\int_a^b x^*(t) x(t)\, dt}$

. $f = t \quad g = t^2 \quad a = 0, b = 1 \qquad \int_0^1 t^3\, dt = \underline{\underline{1/4}}$

$A, B \in \mathbb{R}^{n \times n}$, $\quad \langle A, B\rangle = \text{trace}(B^* A) = \text{trace}(AB^*)$ ✓ Inner product

Norm induced: $\|A\|_F = \sqrt{\text{trace}(A^* A)}$

Let $\mathcal{V}$ be a inner product space and $x, y \in \mathcal{V}$. Then

(a) $|\langle x, y\rangle| \leq \|x\| \|y\|$ (Cauchy – Schwartz inequality)

The equality holds if and only if $x = \alpha y$ for some constant α or $y = 0$.

(b) $\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$ (Parallelogram Law)

(c) $\|x+y\|^2 = \|x\|^2 + \|y\|^2$ if $x \perp y$.

Hilbert space is a complete inner product space with the norm induced by the inner product.

Hilbert space is also a Banach space

e.g. $C[0,1]$, $\|f\|_\infty := \sup_{t \in [0,1]} |f(t)|$ $f(t) \in C[0,1]$ $\longleftarrow$ Banach space complete normed space

$f(t) = 1$, $g(t) = t$ $\|1+t\|_\infty^2 = 4$ $\|1-t\|_\infty^2 = 1$ $4+1=5$

$2\|1\|_\infty^2 + 2\|t\|_\infty^2 = 2+2 = 4 \neq 5$ P.L is not satisfied.

$\therefore$ This is not a Hilbert space.

e.g. $\mathbb{C}^n$ is a finite dimensional Hilbert space.

$\mathbb{C}^{n \times m}$ with $\langle A, B\rangle = \text{trace}(A^* B)$

Hilbert

e.g. $\ell_2(-\infty, \infty)$ consists of all real/complex square summable sequences.

i.e. $(\ldots, x_{-2}, x_{-1}, x_0, x_1, x_2, \ldots)$ $\left(\sum_{i=-\infty}^{\infty} |x_i|^2\right)^{1/2} < \infty$

with inner product $\langle x, y\rangle = \sum_{i=-\infty}^{\infty} x_i^* y_i$ $\forall\ x, y \in \ell_2(-\infty, \infty)$

e.g. $\mathcal{L}_2(J)$: consists of all square integrable and Lebesgue measurable functions defined on the interval $J \subseteq \mathbb{R}$ with the i.p.

$\langle f, g\rangle := \int_J f^* g \, dt$ $\forall\ f, g, \in \mathcal{L}_2(J)$ $\int_J \text{Trace}(f^* g)\, dt$

$$\left.\begin{array}{l} \underline{\underline{\mathcal{L}_2}} = \mathcal{L}_2(-\infty, \infty) \qquad \langle f, g \rangle = \underbrace{\int_{-\infty}^{\infty} f^* g \, dt} \\ \mathcal{L}_{2+} = \mathcal{L}_2[0, \infty) \quad , \quad \mathcal{L}_{2-} = \mathcal{L}_2(-\infty, 0] \end{array}\right\} \text{Spaces of functions}$$